

TrES-1b

R-0223

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_250919 · created 2026-07-17T02:05:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460937.7302 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.049
Transit depth (Rp/Rs)^2	0.00046 +/- 0.021 [%]
Orbital Inclination (inc)	89.95 +/- 1.47
Airmass coefficient 1 (a1)	0.71 +/- 0.53
Airmass coefficient 2 (a2)	0.41 +/- 0.27
Scatter in the residuals of the lightcurve fit is	86.44 %
Best Comparison Star	#2 - [349, 68]
Optimal Aperture	1.74
Optimal Annulus	8.78
Transit Duration (day)	0.0921 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.002 ± 0.049 vs 0.1358 (deviation 1.06σ)
Duration fit vs published	0.0921 d vs 0.1042 d (deviation 0.9σ)
Mid-time O–C	6.6 min
Depth SNR	0.0
Residual scatter	86.44 %

Light curve

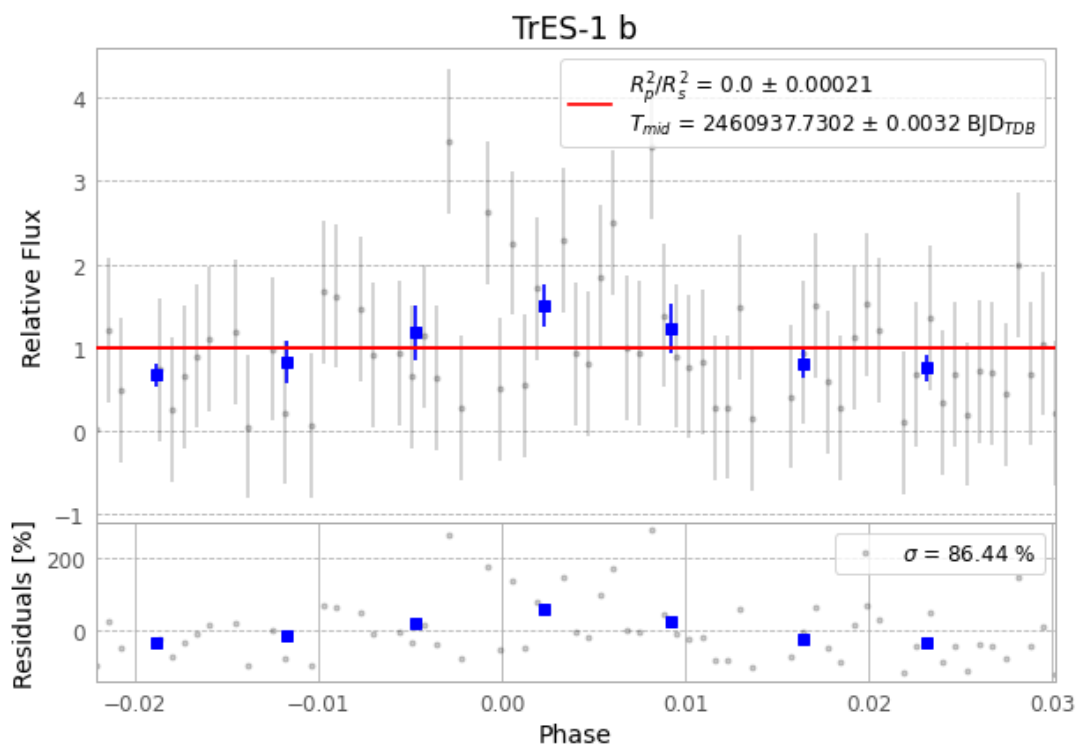


Figure 1 — FinalLightCurve_TrES-1 b_2025-09-18.png (SHA-256 8292471c0953ce64...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [561, 306], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dae9c24cb598025e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

77 FITS frames (51 MB), TRES-1250919035117.FITS → TRES-1250919073915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-250919.log (de59039b1eba...), FinalLightCurve_TrES-1 b_2025-09-18.png (8292471c0953...), FinalLightCurve_TrES-1 b_2025-09-18.csv (47c22680daf9...)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 02:05Z.